

# Generation Interconnection System Impact Study Report

## For Queue Project Q19-1306

Blue Ravine-School 230 kV

February 18, 2021

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1306. The Interconnection Customer proposes a 60.6 MW solar photovoltaic facility interconnecting at the Blue Ravine-School 230 kV substation in the Indiana area. The request was received on October 02, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 436 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$497,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility          | Contingency                          | Loading (%) | Rating (MVA) | DFAX (%) |
|-----------------------------|--------------------------------------|-------------|--------------|----------|
| Murdock - Sabanera 230.0 kV | East Methuen - Henry County 500.0 kV | 104.4       | 731          | 36.8     |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                    | Required Upgrade           | Cost Estimate | Allocation to Project |
|-----------------------------|----------------------------|---------------|-----------------------|
| Murdock - Sabanera 230.0 kV | Reconductor line           | \$17,049,000  | 3% (\$497,000)        |
|                             | Total allocated to project |               | \$497,000             |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.